Fleet Response copilot

An AI first responder for autonomous vehicles that get stuck. It advises; code decides; Mubit turns every step into the fleet's own model.

Mubit Night Shift, London, October 2026
github.com/aswin-giridhar/mubit-fleet

Robotaxis get stuck. Today a person gets them moving.

A double-parked van, roadworks, a police officer waving traffic through, a frightened rider. The car asks a remote agent for context. The agent suggests a lane or a path, and the car stays in control.

Waymo calls this fleet response, and describes it as advice, not control.

~70

remote agents on duty, reported for a fleet of about 3,000 vehicles. Every request an AI resolves correctly is one a person doesn't have to.

Sources: waymo.com/blog/2024/05/fleet-response, waymo.com/blog/shorts/advice-not-control-the-role-of-remote-assistance, TechCrunch Mobility (22 Feb 2026)

A copilot that investigates, advises, and never drives

Triage

Reads messy evidence (sensors, camera caption, rider and staff messages) and names one of ten incident types.

Mubit task 1

→

Respond

Starts with only an alert, calls five tools to investigate, then submits one piece of advice.

Mubit task 2, tool use

→

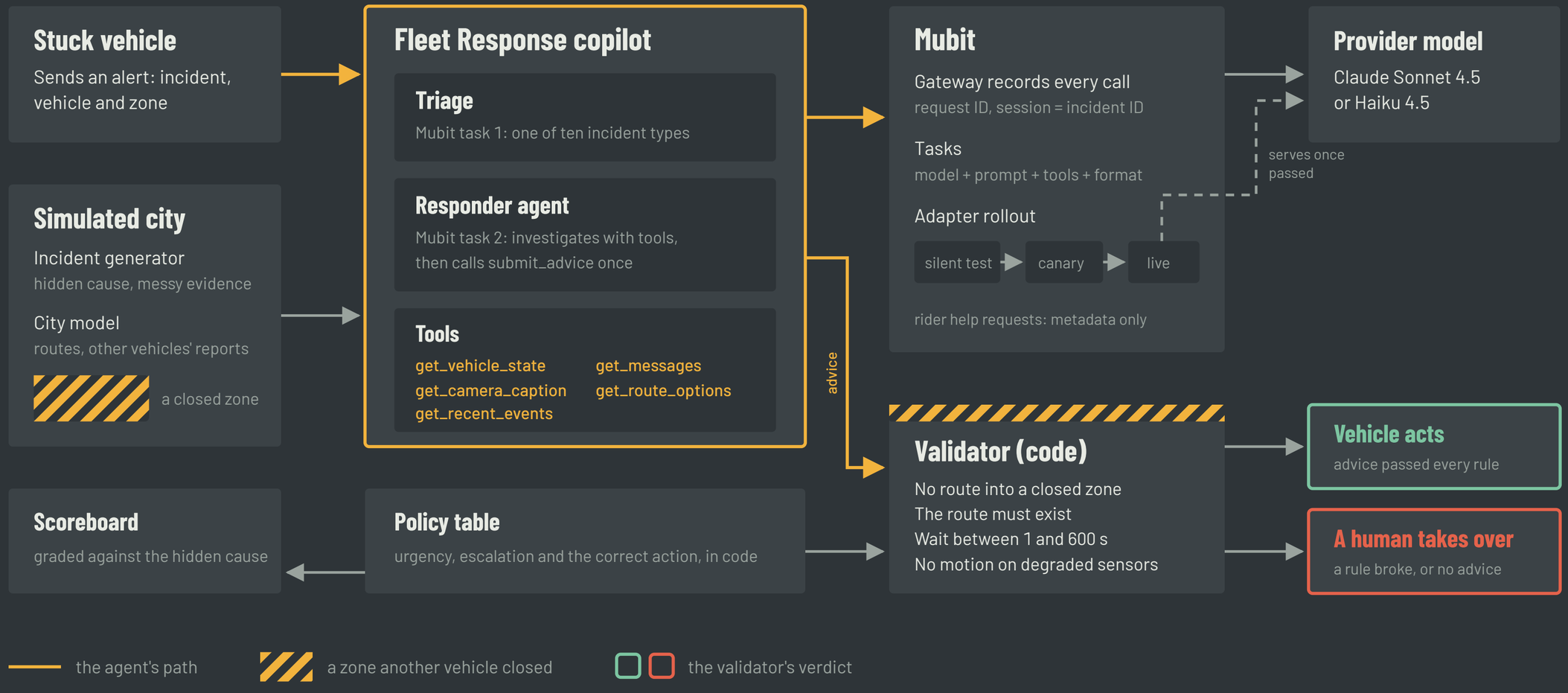
Validate

Code checks the advice against hard safety rules. Anything that breaks one goes to a human.

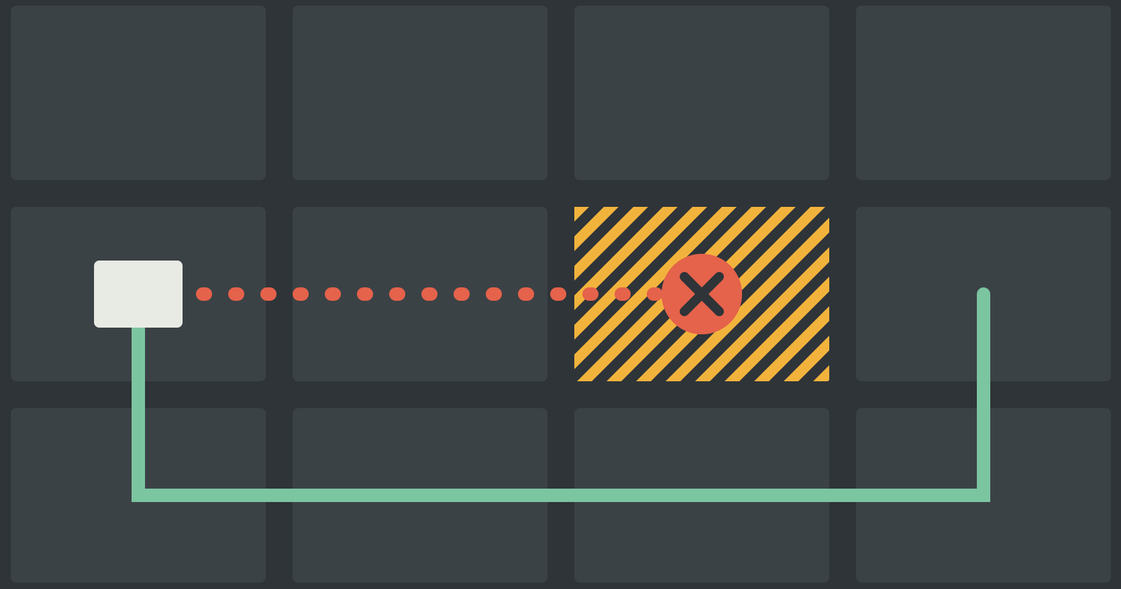
Plain code, no model

How one help request flows

The model proposes. Code decides. Mubit learns a small model for each repeated task.



The model proposes. Code decides.



Fastest route: through a zone another vehicle closed minutes ago. Rejected. The safe route stays open.

No route into a zone another vehicle closed

The route must exist

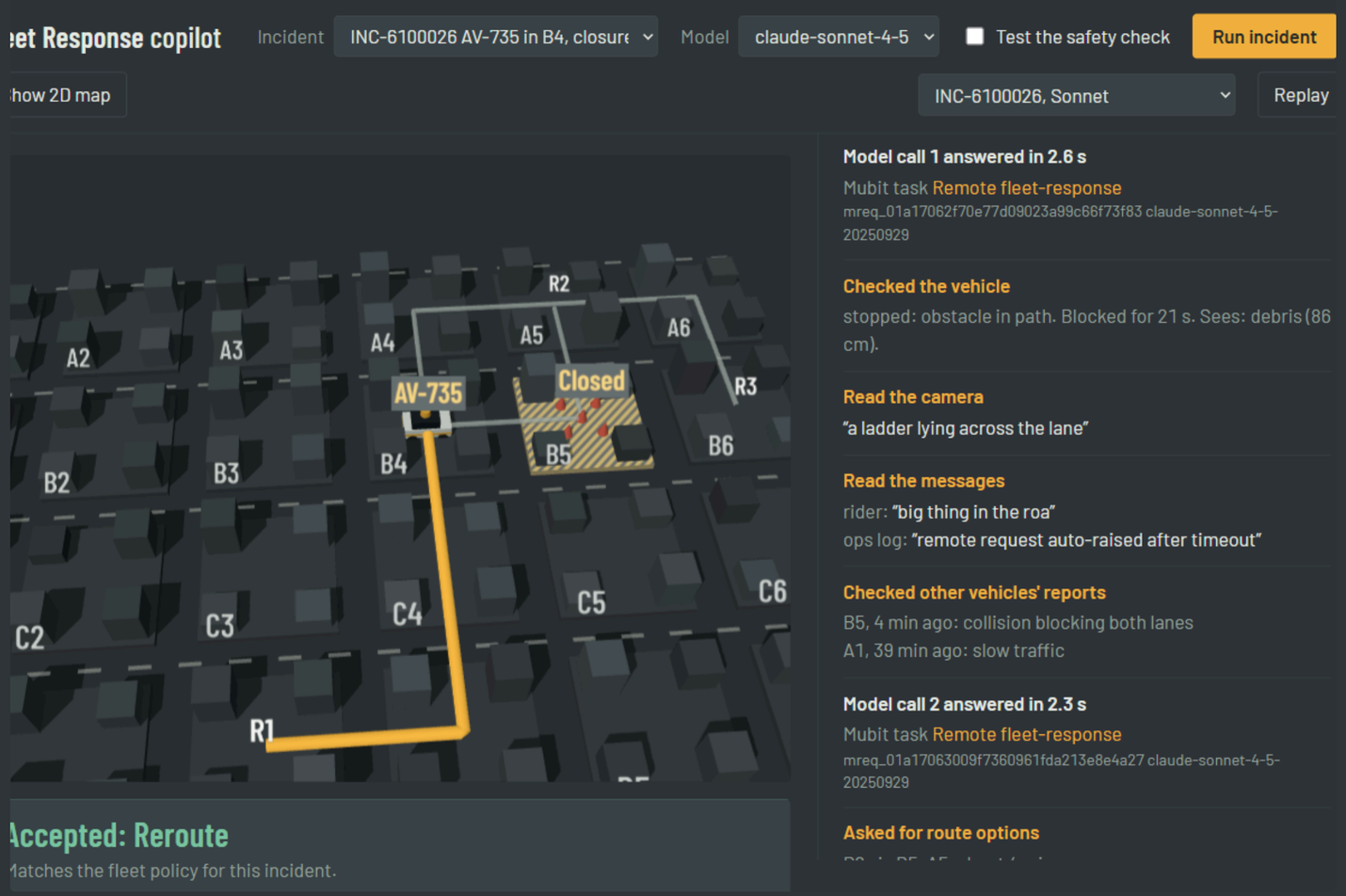
A wait lasts between 1 and 600 seconds

No motion while perception is degraded

Broken rule or no advice: a human takes over

On 40 incidents the validator blocked **1 unsafe suggestion**, from the smaller model. Swapping models cannot make the fleet unsafe.

Live: a vehicle blocked by an empty car



1. An alert arrives: vehicle, zone, nothing else
2. The agent checks the vehicle, camera and messages
3. It reads what other vehicles reported nearby
4. It advises a reroute
5. Code accepts it, or rejects it and calls a human

Every call goes through Mubit

First-line triage

152 calls, one frozen prompt. Adapter v1 passed at 93.8%

Remote fleet-response

254 tool-use calls, one frozen prompt and tool list

25 kept examples — silent test — 10% rollout — live

Automatic once a version answers within 5 points of the provider model. No code change.

Sessions

Each incident is one session, so it never sits on both sides of the train and test split.

Request IDs

Every wrong answer is logged with its Mubit request ID, ready to rate Bad.

Rider privacy

Rider help requests are recorded as metadata only. Mubit never stores or trains on a rider's words.

Served by

Every call logs the model that answered, so the switch to the fleet's own model shows up on its own.

The big model is right more often. The small one is nearly twice as fast.

	Claude Sonnet 4.5	Claude Haiku 4.5
Correct advice	90%	72%
Unsafe advice blocked by code	0	1
Median time to advice	9.5 s	5.3 s
Estimated cost per 1,000 incidents	$17.11	$5.48

40 generated incidents (seed 61), identical for both models, run through Mubit. Costs are estimates from list prices.

That gap is what a distilled model closes

Volume

Help requests grow with the fleet, every vehicle, every day.

Agent loops

About 3,600 input tokens per incident on Sonnet: a tool loop, not a single call.

Waiting costs

A stuck vehicle blocks traffic and a rider the whole time the model thinks.

Ownership

Incident data is safety-sensitive. Each fleet should own the model it trains.

Honest note: triage is easy for every model we tried (both 97%), so it may stay prompt-only. The responder is where a dedicated model earns its place.

What's next

For the copilot

- Roll the triage adapter out: it passed its tests at 93.8%; Mubit's silent test comes next
- Teach the responder: its first version scored 0% on multi-turn tool use
- Rate the logged wrong answers, so outcomes feed back

For Mubit

- A served-by signal on each response, so apps see the switch
- Learnings as fleet memory: what one vehicle learns, every vehicle knows
- A ratings API, so validators can rate replies directly

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